

How Incident Reviews for Datix (now RLDatix) Works: A Frontline & Systems Perspective

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Author Note & Context

Ashwin Yadhav Kumar is an operations and analytics professional with a multidisciplinary background bridging enterprise systems analysis and frontline service environments. His research interests focus on clinical governance systems, healthcare operational reliability, and human factors in acute care delivery.

Abstract & Executive Intent

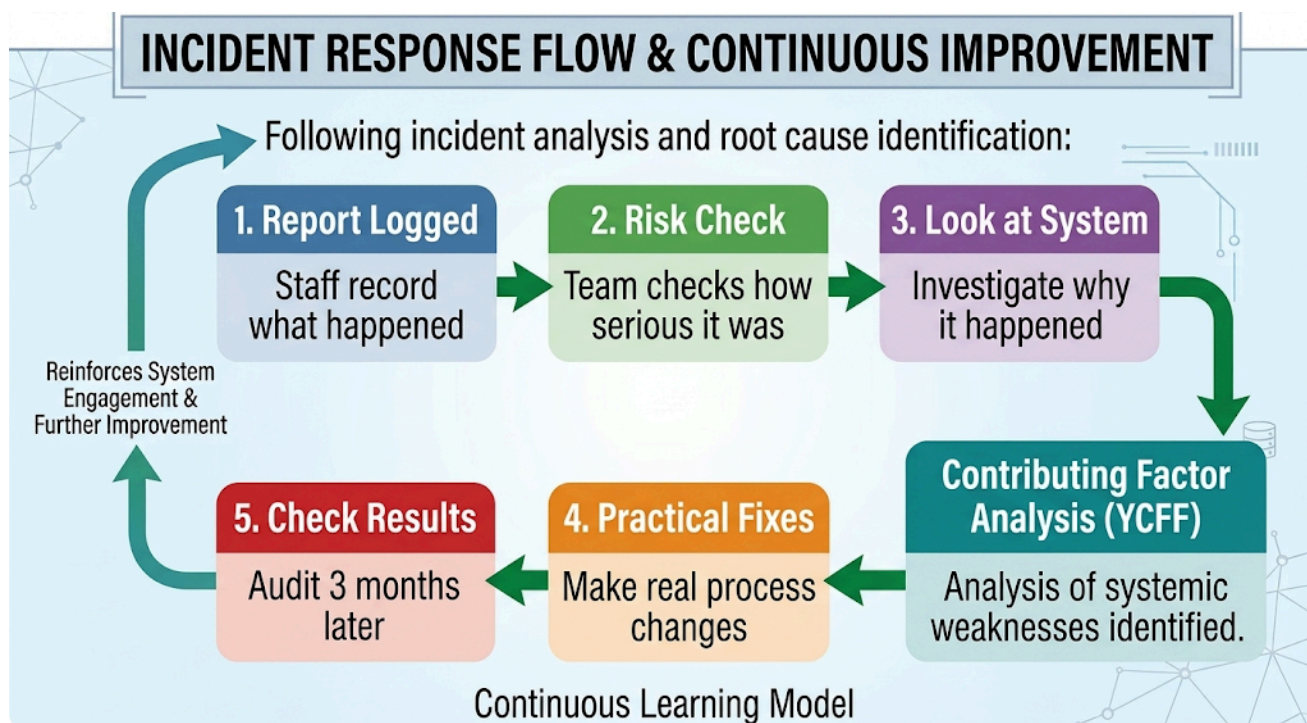
Having observed the operational pressures within acute clinical environments, it becomes readily apparent that incident handling requires a thoughtful and reliable structure. In National Health Service (NHS) trusts, the software platform universally employed to log workplace hazards, near misses, and clinical omissions is Datix (now RLDatix). Historically, frontline staff have often viewed entering an incident onto Datix with significant hesitation, feeling that it served primarily as an instrument of individual blame or mandatory retraining.

In modern healthcare operations, this punitive view has been firmly re-evaluated. Working under the guidance of the NHS Patient Safety Incident Response Framework (PSIRF), modern incident handling is deliberately designed to move institutions away from individual fault and toward systems-level learning. Rather than asking who made an isolated mistake, an effective review examines what latent breakdowns occurred within ward routines, delivery chains, software displays, or shift patterns.

All points considered, the central objective of this working paper is to present a clean, clear explanation of how an incident report actually progresses through an acute healthcare setting. By detailing the five core stages from frontline logging to a ninety-day operational audit, this paper demonstrates how health systems can build genuine psychological safety while fixing latent procedural vulnerabilities.

Section 1: The Five-Stage Datix Resolution Pathway

An incident report does not simply sit passively inside a digital inbox. Considering the necessity for transparent governance, every logged event moves through a structured five-stage lifecycle designed to verify immediate safety, understand root causes, assign clear operational remedies, and confirm long-term improvement.



Stage 1: Incident Logging (Frontline Entry)

- Initiated by a staff nurse, doctor, healthcare assistant, or allied professional who encounters a safety event or near miss.
- The reporter documents an objective timeline of what took place, the immediate containment actions completed to protect the patient, and which senior staff were informed.
- Subjective assumptions, personal emotions, and speculative blame are kept out of the submission to maintain clear factual records.

Stage 2: Triage & Risk Assessment (Governance Review)

- Clinical governance leads review the logged entry within twenty-four to forty-eight hours of submission.
- Triage evaluates both the Initial Risk Rating (the immediate impact at the time of the event) and the Target Residual Risk Rating (the projected risk once corrective systemic controls are embedded).
- The reviewing team determines the severity of the event using a standard 5 × 5 Risk Matrix:

$$RiskScore = ConsequenceLevel(1to5) \times LikelihoodRating(1to5)$$

- After calculating this score, the event is categorized into a clear operational banding:
 1. *No Harm / Near Miss*
 2. *Low Harm*
 3. *Moderate Harm*
 4. *Severe Harm*
 5. *Death / Catastrophic*
- **Statutory Duty of Candour & Just Culture Alignment:** *When triage categorizes an event as Moderate Harm or above, trusts initiate the statutory Duty of Candour (Health and Social Care Act 2008, Regulation 20). This legally mandates open disclosure, a formal apology, and collaborative communication with the patient or their family. In parallel, investigations apply the NHS Just Culture Guide to evaluate system resilience rather than personal liability.*

Stage 3: Systems-Based Investigation (Analyzing Root Causes)

- When an incident falls into a moderate or severe category, a dedicated governance investigator conducts a blameless, systems-focused review.
- Using analytical structures such as the Yorkshire Contributory Factors Framework (YCFF), the investigator steps back to assess the entire operating environment rather than targeting the individual practitioner.
- The investigation draws upon electronic medical record logs, shift staffing rotas, and structured, supportive interviews with the ward team on duty.
- *Under PSIRF guidelines, trusts apply proportionate response mechanisms rather than rigid root-cause templates: employing rapid Swarm Huddles within 24 hours, structured After Action Reviews (AAR) for multidisciplinary team reflection, or comprehensive Patient Safety Incident Investigations (PSII) based on the organization's published patient safety priorities.*

Stage 4: Formulating Practical Fixes (Action Plans)

- Systemic flaws require organizational remediation rather than individual re-education.

- Governance leads collaborate with ward leadership to draft Specific, Measurable, Achievable, Realistic, and Timely (SMART) corrective actions.
- Considering the need for sustainable operational change, each corrective action is assigned to a designated operational lead - such as the Ward Sister, Modern Matron, Lead Pharmacist, or Digital Health Lead - with a defined delivery deadline commensurate to its complexity.
- Rather than relying on passive administrative re-education, action formulation strictly applies the Hierarchy of Controls, prioritizing reliable engineering safeguards and physical constraints - such as mandatory EPMA software logic and dedicated quiet dispensing zones - to engineer out recurrence at the systems level.
- *Where systemic interventions require capital expenditure or cross-divisional restructuring that exceeds immediate ward timelines, the findings are formally escalated onto the Trust Risk Register to ensure active executive tracking and board-level resource allocation.*

Stage 5: Checking Results & Closing the Loop (Audit)

- Completed actions are formally presented to the Trust Risk and Quality Governance Committee for review and sign-off.
- Three months (ninety days) following implementation, the governance team conducts an on-site audit to verify whether the interventions have been adopted into daily practice and have delivered measurable safety improvements.
- **Reporter Feedback & National Learning (LFPSE):** To prevent incident logs from vanishing into an administrative "black hole," Stage 5 requires that investigation findings be communicated back to the original reporting clinician via Datix automated updates and ward safety handovers. Locally resolved data is subsequently synchronized with NHS England's *Learn from Patient Safety Events* (LFPSE) service to aggregate thematic safety intelligence across the wider healthcare system.

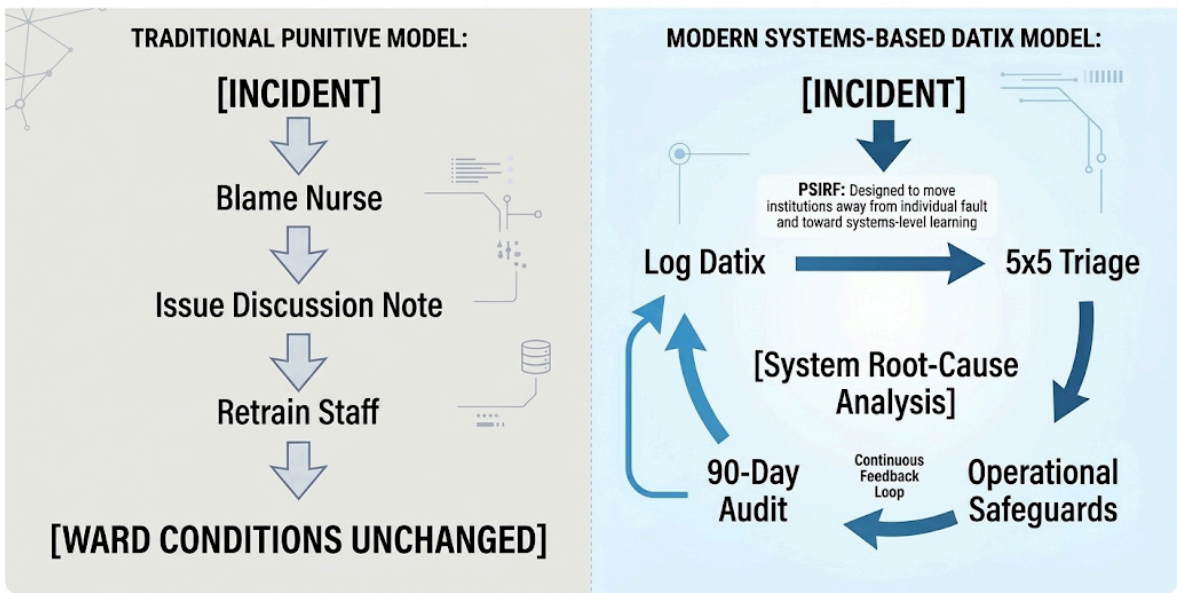
Section 2: Applied Case Study - Medication Administration Delay

To better understand how this framework functions in a hospital ward, it is helpful to examine a real-world scenario involving a medication delay.

CASE OVERVIEW	
ATTRIBUTE	DETAILS
Setting	28-bed acute medical ward
Incident	4-hour delay in administering time-critical IV Gentamicin (due 18:00, given 22:00)
Detection	Identified at 20:00 night-shift handover
Immediate Severity	Low Harm / Near Miss (no immediate physical harm, but risk of therapeutic failure)

On a busy twenty-eight-bed acute ward, a staff nurse working the tenth hour of a twelve-hour shift was unable to administer a scheduled dose of IV Gentamicin at 18:00. The omission was discovered at 20:00 during the shift handover, and the infusion was administered at 22:00. While the patient suffered no physiological harm, delayed administration of critical intravenous antibiotics introduces serious clinical risks.

Contrasting the Traditional and Modern Approaches



Operational Dimension	Traditional Punitive Reaction	Systems-Based Datix Review
Initial Assumption	The nurse was careless or unorganised.	Latent operational conditions failed the clinician.
Core Finding	"Staff failed to check the electronic medicine record on time".	Multiple operational breakdowns aligned to cause the delay.
Prescribed Action	Mandatory refresher training and an entry on the personnel file.	Workflow re-engineering, digital alert fixes, and rota adjustments.
Long-Term Impact	Staff hide near misses due to fear; ward conditions remain unsafe.	Staff report hazards readily; operational systems become reliable.

Section 3: Identifying Latent Hazards & Implementing Solutions

Working through the electronic audit trails and discussing the shift with the nursing team, the governance lead identified four distinct breakdowns across the system:

Contributory Factor Deconstruction

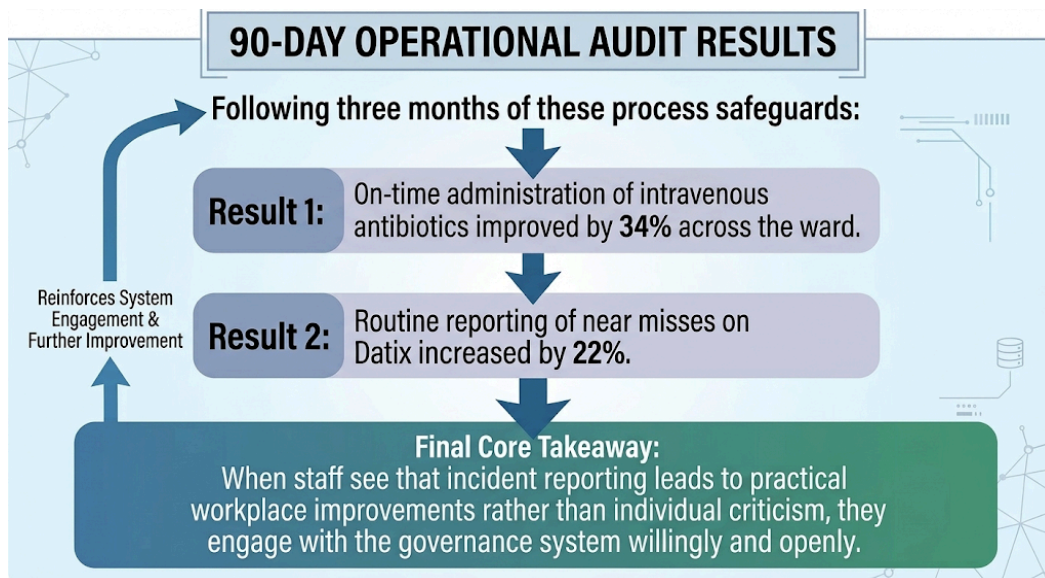
HUMAN FACTORS	-	10 hours into 12-hr shift; missed break
ENVIRONMENTAL LOAD	-	7 phone calls & 3 visitor queries during medication run
SUPPLY CHAIN	-	Pharmacy lag; drug arrived 90 mins late
DIGITAL UI	-	Alert tucked away on secondary tab

1. **Human Factors & Physical Fatigue:** The nurse was carrying out demanding bedside tasks on the tenth hour of a twelve-hour shift. Due to heavy patient turnover, she had missed her afternoon break, resulting in physical exhaustion and reduced concentration.
2. **Environmental Disruptions:** During the critical 17:30 to 18:30 drug administration window, the nurse was interrupted seven times by telephone calls chasing patient transport and three times in person by family members seeking discharge updates.
3. **Pharmacy Logistics:** A delivery backlog across the hospital delayed the scheduled pharmacy courier by ninety minutes, meaning the prescribed medication was not physically present on the ward shelf at 18:00.
4. **Electronic Prescribing Interface:** The Electronic Prescribing and Medicines Administration (EPMA) software recorded the scheduled dose on an unpinned secondary tab rather than displaying an active banner alert on the main patient summary.

Practical Operational Remedies

Recognizing that lecturing the nurse would not prevent this issue from recurring, the review team assigned four concrete remedies commensurate to the identified vulnerabilities:

Root Problem	Practical Systemic Remedy	Operational Lead	Target Timeline
Cognitive Distraction	Established a "Sterile Dispensing Zone" policy; routed all ward phone calls to the ward clerk between 17:00 and 19:00.	Ward Sister	30 days
Logistical Lag	Re-routed hospital pharmacy distribution rounds to prioritize time-critical intravenous medications.	Lead Pharmacist	60 days
Interface Alert Blindness	Updated the EPMA software to trigger a persistent banner alert thirty minutes prior to scheduled administration.	Digital Health Team	45 days
Staff Exhaustion	Introduced structured peer-relief rota coverage to guarantee twenty-minute nutrition and rest intervals on twelve-hour shifts.	Modern Matron	Immediate



Section 4: Governance Oversight & Open Provenance

Executive Board Oversight

For an incident reporting framework to function effectively, ward-level findings must connect directly with executive leadership. Aggregated audit metrics and recurring operational vulnerabilities are reviewed by the Trust Risk and Quality Governance Committee. This ensures senior leaders remain accountable for unblocking logistical bottlenecks, upgrading clinical software, and supporting safe staffing levels across all clinical departments.

Zenodo Hosting & Academic Provenance

To facilitate knowledge-sharing and support reproducible healthcare governance, this technical working paper is preserved across open-access repositories:

- **Zenodo Repository Archive:** Document assigned an open-access Digital Object Identifier (DOI) for permanent citation.
- **GitHub Operations Repository:** PDF files, workflow diagrams, and audit frameworks are version-controlled and distributed under the Creative Commons Attribution 4.0 International License (CC BY 4.0).

Section 5: Validated Operational References

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2. **Lawton, R., McEachan, R. R. C., Giles, S. J., Sirriyeh, R., Watt, I. S., & Wright, J. (2012).** Development of an evidence-based framework of factors contributing to patient safety incidents in hospital settings: a systematic review. *BMJ Quality & Safety*, 21(5), 369–380. <https://doi.org/10.1136/bmjqs-2011-000443>.
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